Samartha H M

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Summary

Electronics and Communication Engineering graduate with hands-on experience in PCB design, real-time firmware development, and hardware-software integration. Proven ability to deliver complete embedded and robotic systems from concept to market within a startup environment. Strong interest in embedded systems, autonomous robotics, and IoT automation.

Education

Srinivas Institute of Technology — **Jun 2026**
B.E. Electronics & Communication Engineering – CGPA: 7.86/10 — *Valachil, Mangalore*

Experience

Experimind Labs Pvt. Ltd. — **Feb 2026 – May 2026**
Core Team Member (Promoted from Electronics Intern within 2 months)

- Spearheaded R&D and full product cycle — ideation, PCB design, prototyping, SMD/through-hole assembly, testing, and market release — for two educational IoT kits delivered to 300+ students across 100+ production units.
- Engineered Kit 1 (14-component sensor board: 11 input sensors, 3 output actuators, custom PCB, JST wiring, battery pack) and Kit 2 (2-motor line follower robot platform with IR/LDR sensing) entirely from scratch with 98%+ first-pass assembly quality.
- Developed embedded control system for Raspberry Pi Pico W-based autonomous land survey robot with waypoint and free-path navigation using GPS, magnetometer, and gyroscope — prototype delivered and demonstrated.

ISPARK Learning Solutions — **Feb – May 2026**
Embedded Systems & IoT Intern (Remote)

- Prototyped AI-powered smart waste segregation system in 5 days: ESP32 (sensors + servo) + ESP32-CAM + Python MobileNetV2 classifier (79% accuracy on 600-image dataset) with custom serial protocol achieving sub-second DRY/WET/SKIP sorting response.
- Implemented complete hardware, firmware, and Python inference engine with confidence thresholding and real-time telemetry dashboard.

Projects

Autonomous Arecanut Drying Robot | *Final Year Capstone Project (Team Lead)*

- Designed 5-motor 4WD platform (±5cm accuracy on 5×5m grid) with dual-controller architecture (Raspberry Pi 5 + Arduino Mega) and custom binary protocol achieving <50ms command latency for reliable real-time teleoperation.
- Configured BTS7960 motor drive circuitry (40A peak load) and implemented grid-based serpentine traversal algorithm ensuring complete autonomous area coverage across agricultural drying yards.
- Addressed post-harvest loss in arecanut farming by replacing labor-intensive manual raking with a fully autonomous platform; established self-hosted Wi-Fi access point enabling smartphone control without internet dependency — critical for rural deployment.
- Orchestrated end-to-end system integration across hardware, real-time firmware, ML pipeline, and React frontend — ensuring all subsystems operated reliably as a single cohesive system under real field conditions.
- Created React dashboard with dark mode, hold-to-move safety mechanism, and real-time telemetry visualization. Led 4-member cross-functional team from design through field deployment and demonstration.

One-Way Vehicle Direction Monitor | *Embedded Safety System*

- Constructed dual-lane wrong-way detection model (95% accuracy) using HC-SR04 sensors with instant servo barrier activation and audio-visual alerts; programmed Arduino Uno with concurrent dual-lane sensor logic.

Personal Projects & Automation Tools | *Independent Builds*

- Shipped SyncPadIO (real-time text sharing), WebsiteON (serverless monitor + Telegram alerts), SonicSync (WebRTC audio), YotuDrive (Reed-Solomon encoding), Mobtola Pro (ADB backup) — leveraging Python, React, Node.js, and TypeScript.

Technical Skills

Programming: Python, C/C++, JavaScript/TypeScript, Arduino IDE
Embedded Systems: Arduino (UNO/Mega), Raspberry Pi (5/Pico W), ESP32, Real-Time Control, PWM, Motor Control, Sensor Integration
Circuit Design & PCB: KiCad, EasyEDA, LTspice, BTS7960 Drivers, Power Electronics, SMD/Through-Hole Soldering, PCB Layout
AI & IoT: MobileNetV2, OpenCV, Computer Vision, IoT System Integration, Real-Time Data Acquisition
Tools & Workflow: Git, VS Code, Linux, Adobe Premiere Pro, Canva, Microsoft Office

Leadership & Extracurriculars

Event Organizer | *ENVISION 2k25 National Tech Fest*

- Coordinated Line Follower robotics competition for 100+ participants and led promotional design team.

Editor & Designer | *STEADY Department Club (2024 & 2025)*

- Published 50-page technical magazine distributed to 500+ students and faculty. Managed 50+ technical and non-technical college events.

Active Member & Junior Mentor | *TECHBOTS Robotics Club*

- Mentored junior students in embedded programming and PCB design. Supported robotics systems development including line-following and security automation.

Content Creator | *DreamBeatz YouTube Channel*

- Music and lyrical edit content with 3K+ subscribers.